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TWO FAMILIES, ONE CONJECTURE

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[1] EACH FAMILY ALONE – where it works and where it fails

n	prime?	residue	gap	best	true floor	verdict
5	True	None	3	3	3	attained by gap
6	False	3	3	3	3	attained by residue & gap
7	True	None	3	3	3	attained by gap
8	False	4	4	4	4	attained by residue & gap
9	False	3	4	3	3	attained by residue
10	False	4	4	4	4	attained by residue & gap
11	True	None	4	4	4	attained by gap
12	False	4	4	4	4	attained by residue & gap
14	False	4	5	4	4	attained by residue
16	False	4	5	4	4	attained by residue

agreements 10 disagreements 0

Look at the pattern rather than the totals:

primes 5, 7, 11 residue is EMPTY, gap hits the floor exactly
n = 9, 14, 16 gap MISSES, residue hits it exactly

Neither family alone suffices. Together they have not missed once.

[2] THE CONJECTURE

For every n, the minimum sensitivity of a non-constant cyclically invariant Boolean function is attained by either

 a RESIDUE function (depends only on counts of ones per class mod d)
or a GAP function (depends only on the multiset of spacings)

Both families are elementary, and both are exponentially smaller than the space of all cyclically invariant functions. At n = 25 the whole space has 2¹³⁴²²⁰⁸ members; the residue family has 1,560 profiles.

[3] FALSIFIABLE PREDICTIONS

The conjecture pins the two floors our exhaustive solver could not reach. Both are checkable by anyone with more compute.

n	proven bracket	residue	gap	PREDICTION
13	[4, 5]	None	5	5
15	[4, 5]	5	5	5

$n = 13$ is prime, so only the gap family applies $\rightarrow$ predicts 5.

$n = 15 = 3 \times 5$, both apply, both give 5 $\rightarrow$ predicts 5.

We PROVED sensitivity 3 is impossible at both. So each prediction reduces to a single question: does a sensitivity-4 function exist? The conjecture says no, in both cases. One counterexample kills it.

[4] AND THE SQUARE LAW SURVIVES INSIDE IT

n	sqrt(n)	residue	gap	best
4	2	None	None	--
9	3	3	4	3
16	4	4	5	4
25	5	5	None	5
36	6	6	None	6

At every perfect square the residue family gives exactly $\text{sqrt}(n)$, and where the gap family is also known it is strictly worse. So squares are residue territory, primes are gap territory, and the floor is the better of the two.

WHAT WOULD BREAK THIS

- * any n where the true floor is strictly below BOTH family optima
- * a sensitivity-4 cyclically invariant function on 13 or 15 bits
- * a third natural family beating both somewhere

All three are ordinary computations for someone with more machine time than this session had. That is the point of writing it this way.